Complete Seaborn Graphs Cheat Sheet

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1. Scatter Plot

Use Case: Relationship between two numerical variables

Syntax:

```
sns.scatterplot(x="total_bill", y="tip", data=df)
```

2. Line Plot

Use Case: Trend analysis

Syntax:

```
sns.lineplot(x="size", y="total_bill", data=df)
```

3. Bar Plot

Use Case: Compare categories

Syntax:

```
sns.barplot(x="day", y="total_bill", data=df)
```

4. Count Plot

Use Case: Count frequency of categories

Syntax:

```
sns.countplot(x="day", data=df)
```

5. Histogram

Use Case: Distribution analysis

Syntax:

```
sns.histplot(df["total_bill"], kde=True)
```

6. KDE Plot

Use Case: Smooth probability distribution

Syntax:

```
sns.kdeplot(df["total_bill"])
```

7. Rug Plot

Use Case: Show observations

Syntax:

```
sns.rugplot(df["total_bill"])
```

8. Box Plot

Use Case: Outlier detection

Syntax:

```
sns.boxplot(x="day", y="total_bill", data=df)
```

9. Violin Plot

Use Case: Density + quartiles

Syntax:

```
sns.violinplot(x="day", y="total_bill", data=df)
```

10. Strip Plot

Use Case: Scatter points for categories

Syntax:

```
sns.stripplot(x="day", y="total_bill", data=df)
```

11. Swarm Plot

Use Case: Better categorical scatter

Syntax:

```
sns.swarmplot(x="day", y="total_bill", data=df)
```

12. Boxen Plot

Use Case: Large dataset analysis

Syntax:

```
sns.boxenplot(x="day", y="total_bill", data=df)
```

13. Point Plot

Use Case: Category averages

Syntax:

```
sns.pointplot(x="day", y="total_bill", data=df)
```

14. Pair Plot

Use Case: Relationships among all numerical columns

Syntax:

```
sns.pairplot(df)
```

15. Joint Plot

Use Case: Bivariate analysis

Syntax:

```
sns.jointplot(x="total_bill", y="tip", data=df)
```

16. Heatmap

Use Case: Correlation matrix

Syntax:

```
sns.heatmap(corr, annot=True)
```

17. Clustermap

Use Case: Hierarchical clustering

Syntax:

```
sns.clustermap(corr)
```

18. Facet Grid

Use Case: Multiple plots by category

Syntax:

```
g = sns.FacetGrid(df, col="sex")
```

19. Pair Grid

Use Case: Custom pairwise visualization

Syntax:

```
g = sns.PairGrid(df)
```

20. Rel Plot

Use Case: Relational plots

Syntax:

```
sns.relplot(x="total_bill", y="tip", data=df)
```

21. Cat Plot

Use Case: Categorical visualization

Syntax:

```
sns.catplot(x="day", y="total_bill", kind="bar", data=df)
```

22. Dis Plot

Use Case: Distribution visualization

Syntax:

```
sns.displot(df["total_bill"])
```

23. Reg Plot

Use Case: Regression analysis

Syntax:

```
sns.regplot(x="total_bill", y="tip", data=df)
```

24. Resid Plot

Use Case: Residual analysis

Syntax:

```
sns.residplot(x="total_bill", y="tip", data=df)
```

25. ECDF Plot

Use Case: Cumulative distribution

Syntax:

```
sns.ecdfplot(data=df, x="total_bill")
```

26. lmplot

Use Case: Regression with categories

Syntax:

```
sns.lmplot(x="total_bill", y="tip", hue="sex", data=df)
```

27. Hexbin Plot

Use Case: Dense scatter visualization

Syntax:

```
plt.hexbin(df["total_bill"], df["tip"], gridsize=20)
```

28. Pie Chart

Use Case: Percentage contribution

Syntax:

```
df["day"].value_counts().plot.pie(autopct="%1.1f%%")
```

29. Dist Plot (Deprecated)

Use Case: Distribution analysis

Syntax:

```
sns.distplot(df["total_bill"])
```

Important Seaborn Settings

Themes:

```
sns.set_style("darkgrid")
```

Palettes:

```
sns.set_palette("deep")
```

Figure Size:

```
plt.figure(figsize=(10,5))
```

Titles and Labels:

```
plt.title("Sales Analysis")
plt.xlabel("Month")
plt.ylabel("Sales")
```